FFGFT as a Test Instance for HLV: the Shared φ-Icosahedral Object

Why the common geometric object and the E_0 asymmetry open a test direction FFGFT→HLV — without endorsing HLV's formulations

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.1 Question: the same object — what HLV does with it stays HLV's business

One statement stands at the outset and is meant to be unambiguous: FFGFT and HLV work with the *same* geometric object. This is not an interpretation and not a thesis, but a computable fact, established below — the φ-icosahedral axis system on which $\theta = 2/9$ sits as an invariant. Everything else in this document is a consequence of this single finding.

Strictly to be separated from it is a second question, which is here *not* asked and *not* answered: what HLV does with this object — which vortex or edge-mode readings, which derivations and which formulations HLV builds upon it. That is HLV's business, not FFGFT's. Whether those formulations are correct is here neither claimed nor tested nor addressed. FFGFT establishes the shared object and — via E_0 — provides a test direction; for the use HLV makes of the object, HLV alone is responsible.

The purpose of the document is thereby deliberately two-fold and modest. It records that both compute over the same object, and that FFGFT offers a way to measure HLV against the empirical data; and it draws at the same time the boundary beyond which nothing is asserted about HLV.

.2 The shared φ-icosahedral object

In the available computational representation of HLV — the cut-and-project of the homology check script `ffgft_hlv_homology_check_v3.py` — physical space is generated by the icosahedral projector with parameter $\tau = \varphi$, $\varphi = \frac{1+\sqrt{5}}{2}$. Its six projected axes are the six five-fold axes of the icosahedron: all fifteen pairwise angles equal $\arccos(1/\sqrt{5}) = 63{,}435°$. On exactly these axes lives FFGFT's C_3-in-A_5 doubling (Dok. 285, Dok. 293).

It is therefore not two similar structures but the same object. Applying to these axes the five-fold redistribution of the C_3 modes (Dok. 293), the share into the trivial mode is exactly $p_0 = 2/9$. The Koide angle $\theta = 2/9$ is thereby an *invariant on the shared object*, not a translation of one framework into the other. The value is moreover specific to the golden ratio: only $\tau = \varphi$ makes the angle spectrum exactly uniform and the trivial share exactly $2/9$; the silver value $\tau = 1 + \sqrt{2}$ gives $0{,}2534$ and an already measurably shifted spectrum (angles $45°$ and $69{,}3°$). The shared object is thus not just any cut-and-project, but the φ-icosahedral one.

This statement is computational and free of circularity: it follows from the comparison of the projector axes and the mode redistribution, checkable in `ikosaeder_theta_delta.py` (seed 20780458) and in the reproducible angle-spectrum check `a5_bridge_discrimination.py`.

.3 The E_0 asymmetry

The shared object is scale-free. The geometry knows neither mass nor energy; it yields pure numbers — the angle and the denominator-9 weights (Dok. 293). Here the actual asymmetry between the two frameworks sets in.

FFGFT possesses a reference point E_0 (Dok. 292) that binds the scale-free pattern to the measured leptons e, μ, τ and fixes the absolute position. FFGFT can thus *compute back empirically* from the pattern: from the shared object, the denominator-9 weights and E_0, absolute masses follow. HLV possesses no such scale. It carries the pattern — the shared geometric object — but no anchor in energy or mass. HLV can therefore obtain no masses from the object alone. It can only do so if it adopts E_0 from FFGFT; in that case, however, it is FFGFT's empirical content that computes, merely expressed in HLV's coordinates.

.4 The test direction: from FFGFT to HLV

From the shared geometry and the one-sided scale a directed relation follows. Because FFGFT supplies the missing scale, FFGFT offers a way to test the geometric content of HLV against the data: one equips the shared object with E_0 and tests whether it reproduces the lepton masses. This is a verification direction from FFGFT to HLV.

It is expressly *not* a confirmation that HLV's own formulations are correct. The shared object states only that both use the same geometry — not that the interpretations and derivations HLV erects on this geometry are valid. The test direction tests the correspondence of geometry and masses with FFGFT's means; it does not touch HLV's own construction and does not ennoble it.

Conversely: HLV supplies FFGFT empirically with nothing it does not already have. The pattern is shared, the scale FFGFT has itself. There is no empirical transfer from HLV to FFGFT, and FFGFT does not need HLV for the masses. The empirical arrow is one-sided.

.5 What the finding says and what it does not

Established and computationally secured is: the shared φ-icosahedral object; $\theta = 2/9$ as its invariant; the specificity for $\tau = \varphi$; and the E_0 asymmetry, from which the test direction follows.

Not established and not claimed is: the correctness of the formulations HLV contains; an independent empirical confirmation of FFGFT by HLV; and the question of whether the icosahedral projector is *intrinsic* to HLV's fundamental claim or constitutes a modelling choice in the check scripts. The latter is for HLV itself to clarify and is here left open.

.6 Methodological placement

Within FFGFT's three methodological layers the finding places clearly. The shared object is a bridge finding: a geometric, algebraically demonstrable correspondence. The back-computation to masses is a core derivation, it follows from ξ and E_0 and is FFGFT-internal. HLV's contribution to the empirical content is zero. In the sense of P35 the statement that HLV is correct remains a candidate and not a derivation; derived is the shared object alone.

.7 The first test is superseded: what it reduces to

The first check attempt — the homology check `ffgft_hlv_homology_check_v3.py` — was a topological point-cloud diagnostic. It projected the cut-and-project into a three-dimensional point cloud, built a kNN filtration over it and measured the persistent H_1 homology, HLV against crystalline and quasicrystalline controls. This approach is metric- and k-dependent, bound to a finite cloud, and lossy in the sense that it sees the structure only through a coarse topological shadow.

With the genuine Hilbert-space variant formulated in Dok. 293 — on the line of Dok. 230/282 — this first test is superseded. The same structure is there exactly and losslessly: as C_3 Fourier modes on the $\mathbb{C}^3$ fibre, as a Z_3 circulant, with the A_5 embedding via R_5; and $\theta = 2/9$ appears as the exact spectral invariant $p_0 = |\langle v_0 | R_5 v_e \rangle|^2 = 2/9$, computed directly rather than estimated over a noisy topology.

The first test thereby reduces to a *coarse quasiperiodicity detector*. It can certify that the HLV cloud is quasiperiodic and not crystalline — and nothing finer. For the actually load-bearing distinction it is structurally blind: the persistent H_1 of a point cloud does not see the axis-angle structure on which $2/9$ hangs. The golden value $\tau = \varphi$ (exactly $2/9$) and the control value $\tau = \sqrt{2}$ ($0{,}2139$) produce topologically nearly indistinguishable clouds; the difference lives in the angle spectrum, which the homology does not resolve.

Two consequences follow. First, the earlier finding of a "only weak separation from generic" is not a statement about φ or A_5, but the expectable blindness of the coarse statistic — additionally aggravated by the fact that the "generic" control used was itself built from the icosahedral projector with $\tau = \sqrt{2}$, and its angle spectrum ($61{,}87^\circ$ and $70{,}53^\circ$) is structured and not scattered. Second, the remedy for this blindness is not a cleaner control for the coarse test, but the exact spectral invariant of the Hilbert-space variant itself. It resolves exactly what the point-cloud topology was blind to.

The shared-object finding therefore rests on the exact Hilbert-space formulation, not on the superseded homology diagnostic. The φ-specificity of $\theta = 2/9$ is visible only in the spectral variant.

.8 The second test: what it resolves and where it becomes circular

The second check attempt — the instrument `a5_bridge_discrimination.py` (seed 20780458) — uses exactly the statistic the first lacked: the angle spectrum of the six projected axes, measured as distance to the icosahedral reference spectrum in which all fifteen pairwise angles equal $63{,}435^\circ$. It thereby resolves the axis-angle structure on which $2/9$ hangs. It is calibrated so that an exactly icosahedral object gives distance zero, while generic random tight frames and a truly generic $\sqrt{2}$ construction lie in a broad generic range. It is thus the instrument that can see φ and A_5 where the first was blind.

What the second test yields on the available representation is two-fold. First, the HLV projector with $\tau = \varphi$ reads exactly icosahedral: all fifteen angles $63{,}435^\circ$, distance zero, and the five-fold redistribution on its axes gives exactly $2/9$. This, however, is a consequence of the construction — the object *is* the icosahedral projector — so the reading is circular and no independent confirmation of A_5. The instrument correctly treats this not as a verdict but makes the circularity explicit. Second, it demonstrates quantitatively that the earlier "generic" control with $\tau = \sqrt{2}$ is not generic: RMS distance $3{,}47$ to the icosahedral reference spectrum (the exactly icosahedral object has distance 0), a two-valued spectrum $61{,}87^\circ/70{,}53^\circ$, and a redistribution weight $0{,}2139$ that is not $2/9$. This distance lies far below the generic band of random frames (median RMS about 22, 5–$95\,\%$ roughly 17–28): the $\sqrt{2}$ control is thus structured and not generic. The φ-specificity is thereby nailed down: only $\tau = \varphi$ makes the spectrum exactly uniform and the weight exactly $2/9$.

The yield of the second test is thus diagnostic and instrumental, not confirming. It provides the correct, reproducible statistic that resolves what the first could not; it quantifies the circularity of a check of the available object and the contamination of the earlier control; and it isolates φ as the distinguished value. What it does not yet provide is an independent confirmation: that would require an HLV object represented *without* a baked-in icosahedral projector. Until such an object is available, the instrument stands ready, and the shared-object finding rests on the exact spectral invariant, not on a non-circular discrimination.

.9 Status

Secured is the shared φ-icosahedral object and the E_0 asymmetry, and from it a test direction from FFGFT to HLV: FFGFT offers the possibility of testing the geometric content of HLV against the measured masses. No correctness of the formulations HLV contains is claimed. One point remains open: whether the icosahedral projector is intrinsic to HLV. Bound to it is the only still possible independent check — a non-circular discrimination requiring an HLV object without a baked-in icosahedral projector; the instrument `a5_bridge_discrimination.py` stands ready for it. The earlier contrast via persistent homology is by contrast settled, not open — it is superseded by the exact Hilbert-space variant and was structurally blind to the φ question anyway.

$\theta = 2/9$ itself stands independent of all this. It follows from the Z_3 circulant diagonalisation and the icosahedral redistribution (Dok. 293) and needs HLV in no way. The relation recorded here is a shared geometry with a one-sided test direction, not a mutual confirmation.